

Optimization TA 3

Kacper Krasowski

Exercise 3

Problem

Let (x^n) be a monotone sequence in $\mathbb{R}$.

Show that it converges if and only if it is bounded.

Hint: Show that it converges to $\sup\{x^n\}$.

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Let (x^n) be a **monotone** sequence in $\mathbb{R}$.

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Hint: Show that it converges to $\sup\{x^n\}$.

What we actually need to show

Convergent and Monotone $\Rightarrow$ Bounded (Already done in Ex1)

Bounded and Monotone in $\mathbb{R} \Rightarrow$ Convergent.

Exercise 3: Definitions

Bounded and Monotone in $\mathbb{R} \Rightarrow$ Convergent.

Definitions

Bounded: (x^n) is bounded if there exists $M > 0$ such that $|x^n| \leq M$ for all n .

Monotone: A sequence (x^n) is monotone if it is either decreasing or increasing.

Convergent: A sequence (x^n) converges to x if, for every $\varepsilon > 0$, there exists N such that $|x^n - x| < \varepsilon$ for all $n \geq N$.

Supremum: The supremum of a set S is its smallest upper bound.

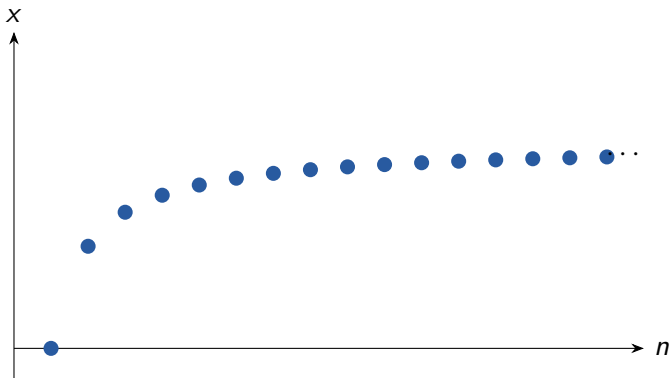
Exercise 3: Sketch of the Proof

Bounded and Monotone in $\mathbb{R} \Rightarrow$ Convergent.

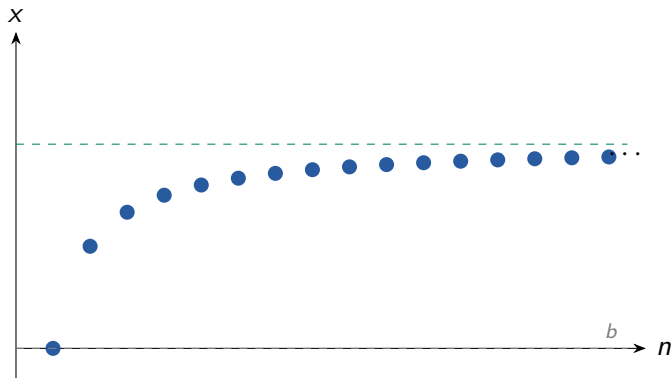
1. Define $d := \sup\{x^n\}$.
2. Since d is the least upper bound and x^n is monotonic, then $d - \epsilon < x^n$ ($n \geq N$).
3. Rearrange and get the definition of convergence.

(This holds only for increasing but the argument for decreasing is symmetrical with infimum instead of supremum)

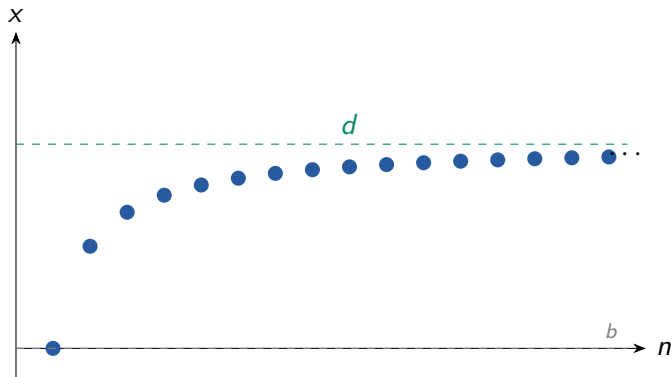
Exercise 3: Approaching the least upper bound



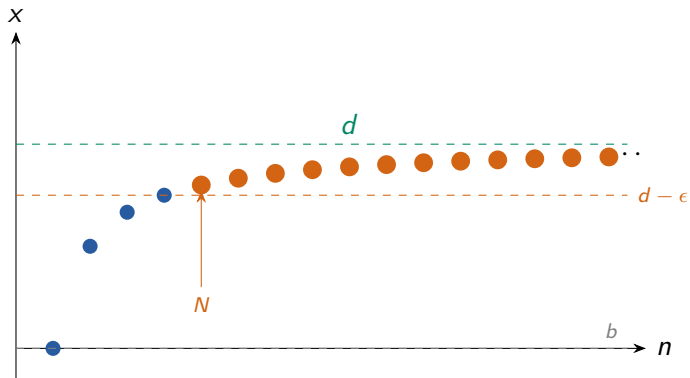
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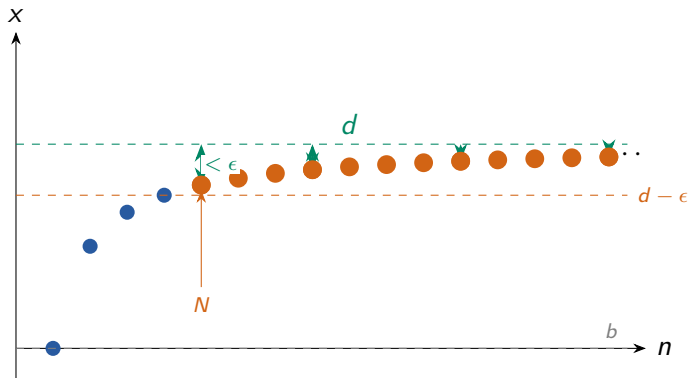
Exercise 3: Approaching the least upper bound



Exercise 3: Approaching the least upper bound



Exercise 3: Approaching the least upper bound



Exercise 6

Problem

In this exercise, you will show that every sequence (x^n) in $\mathbb{R}$ has a monotone subsequence.

To do so, call an index $n \in \mathbb{N}$ a **peak** of the sequence if $x^n > x^m$ for all $m > n$. That is, following x^n , all elements are less than x^n . If the sequence has more than one peak, denote them by $n_1 < n_2 < \dots$.

(a) Show that if the number of peaks is infinite, the subsequence (x^{n_i}) is decreasing.

(b) Finally, show that if the number of peaks is finite, there must exist an increasing subsequence.

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(a) Infinite peaks $\Rightarrow (x^n)$ has a Decreasing subsequence.

(b) Finite peaks $\Rightarrow (x^n)$ has an Increasing subsequence.

Exercise 6: Definitions

(a) Infinite peaks $\Rightarrow (x^n)$ has a Decreasing subsequence.

(b) Finite peaks $\Rightarrow (x^n)$ has an Increasing subsequence.

Definitions

Subsequence: A sequence (x^{n_k}) is a subsequence of (x^n) if

$$n_1 < n_2 < n_3 < \dots$$

and each x^{n_k} is an element of the original sequence.

Infinite sequence: A sequence (x^n) is infinite if it is not finite.

Finite sequence:

A sequence (x^n) is finite if it contains only finitely many elements.

Exercise 6a: Sketch of Proof

(a) Infinite peaks $\Rightarrow (x^n)$ has a Decreasing subsequence.

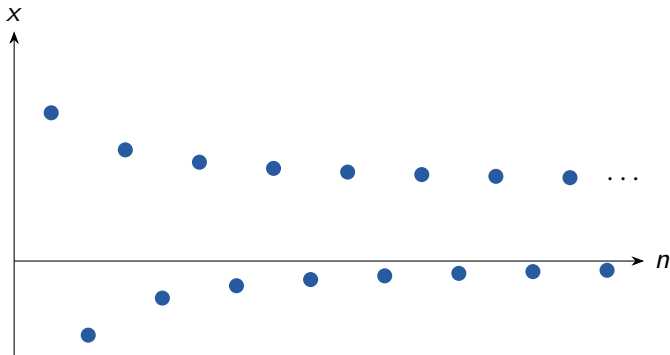
1. Note that peaks themselves as a subsequence prove this.

Exercise 6b: Sketch of Proof

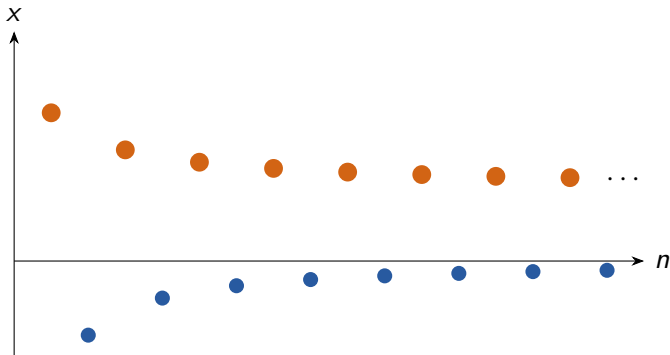
(b) Finite peaks $\Rightarrow (x^n)$ has an Increasing subsequence.

1. Since peaks are finite, then there must be a last one.
2. Next element after the last peak must have a greater or equal element.
3. This greater or equal element must also have a greater or equal element.
4. Repeat 3.

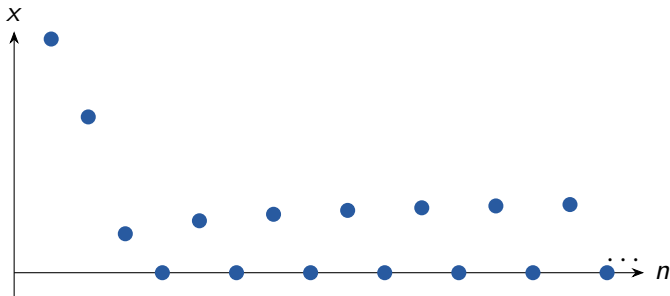
Exercise 6a: Infinitely many peaks



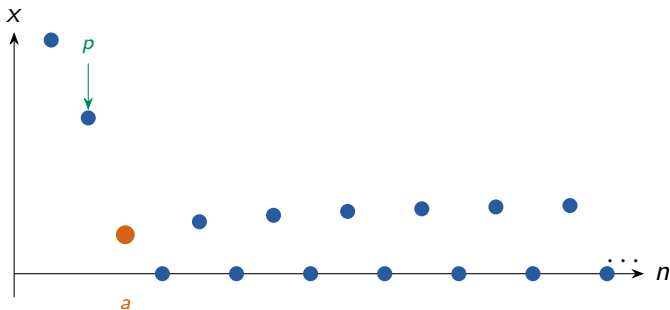
Exercise 6a: Infinitely many peaks



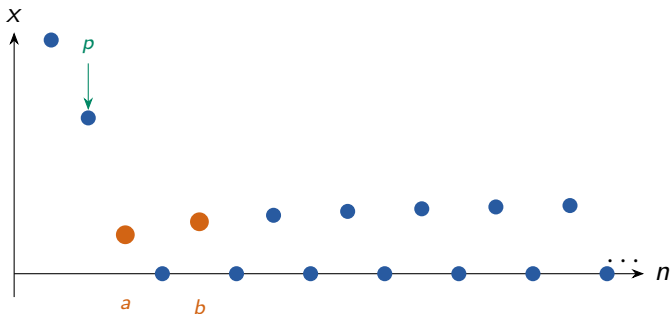
Exercise 6b: Beyond the last peak



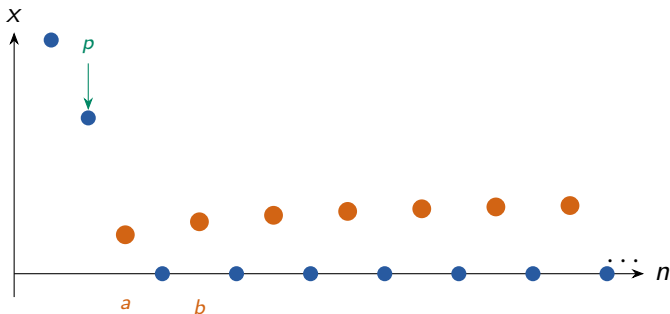
Exercise 6b: Beyond the last peak



Exercise 6b: Beyond the last peak



Exercise 6b: Beyond the last peak



Exercise 7

Problem

(Bolzano-Weierstrass theorem) Use the previous result to prove that every bounded sequence on $\mathbb{R}$ has a convergent subsequence.

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(**Bolzano-Weierstrass theorem**) Use the previous result to prove that every **bounded** sequence on $\mathbb{R}$ has a **convergent subsequence**.

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Bounded in $\mathbb{R} \Rightarrow (x^n)$ has a Convergent subsequence

Exercise 7: Definitions

Bounded in $\mathbb{R} \Rightarrow (x^n)$ has a Convergent subsequence

Definitions

Bounded: (x^n) is bounded if there exists $M > 0$ such that $|x^n| \leq M$ for all n .

Monotone: A sequence (x^n) is monotone if it is either decreasing or increasing.

Convergent: A sequence (x^n) converges to x if, for every $\varepsilon > 0$, there exists N such that $|x^n - x| < \varepsilon$ for all $n \geq N$.

Subsequence: A sequence (x^{n_k}) is a subsequence of (x^n) if

$$n_1 < n_2 < n_3 < \cdots$$

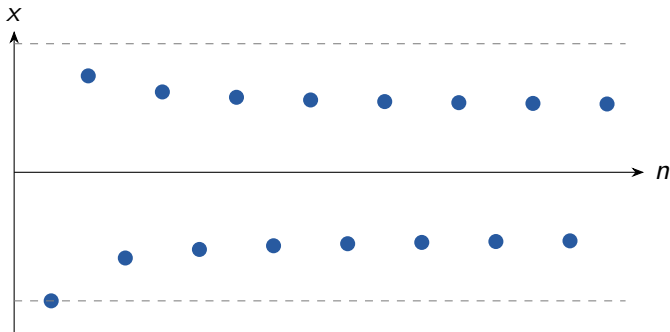
and each x^{n_k} is an element of the original sequence.

Exercise 7: Sketch of Proof

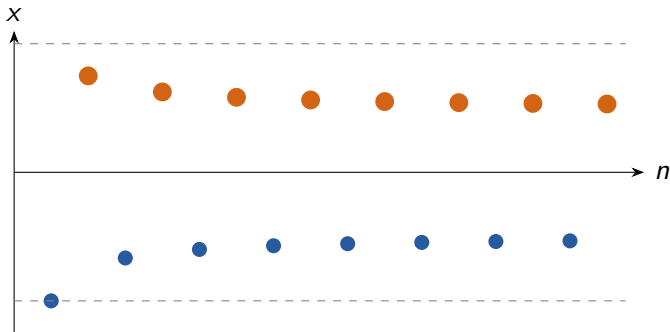
Bounded in $\mathbb{R} \Rightarrow (x^n)$ has a Convergent subsequence

1. By Exercise 6: Every x^n has a monotone subsequence in $\mathbb{R}$.
2. If a sequence is bounded then its subsequences are also bounded.
3. By Exercise 3: Every monotone and bounded sequence in $\mathbb{R}$ is convergent.

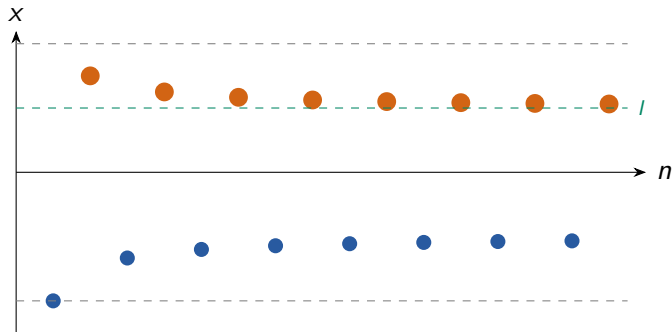
Exercise 7: Extract, keep the bounds, take the limit



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Exercise 8

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Use the previous result to show that $\mathbb{R}$ is complete. That is, show that every Cauchy sequence on $\mathbb{R}$ converges.

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Every Cauchy sequence on $\mathbb{R}$ converges.

Exercise 8: Definitions

Every Cauchy sequence on $\mathbb{R}$ converges.

Definitions & Theorems

Cauchy sequence: A sequence (x^n) is Cauchy if, for every $\varepsilon > 0$, there exists N such that $|x^n - x^m| < \varepsilon$ for all $n, m \geq N$.

Bounded: (x^n) is bounded if there exists $M > 0$ such that $|x^n| \leq M$ for all n .

Bolzano–Weierstrass Theorem: Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Convergent: A sequence (x^n) converges to x if, for every $\varepsilon > 0$, there exists N such that $|x^n - x| < \varepsilon$ for all $n \geq N$.

Exercise 8

Every Cauchy sequence on $\mathbb{R}$ converges.

1. Show that Cauchy sequences are Bounded in $\mathbb{R}$.
 - 1.1 Triangle inequality for $n, m > N$
 - 1.2 Rest by Finiteness
2. Use Bolzano–Weierstrass Theorem to show that Cauchy sequences have convergent subsequences.
3. Use the properties of Cauchy sequences to show that they converge given their subsequences converge.
 - 3.1 Cauchy's subsequence inherits the property.
 - 3.2 Cauchy subsequence is convergent.
 - 3.3 Triangle inequality.

Exercise 1

Problem

Show that a Cauchy sequence is convergent if and only if it has a convergent subsequence.

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Show that a **Cauchy** sequence is **convergent** if and only if it has a **convergent subsequence**.

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Cauchy and Convergent $\Rightarrow$ Convergent Subsequence

Cauchy and Convergent Subsequence $\Rightarrow$ Convergent (Done in Ex8)

Exercise 1: Definitions

Cauchy and Convergent $\Rightarrow$ Convergent Subsequence

Definitions

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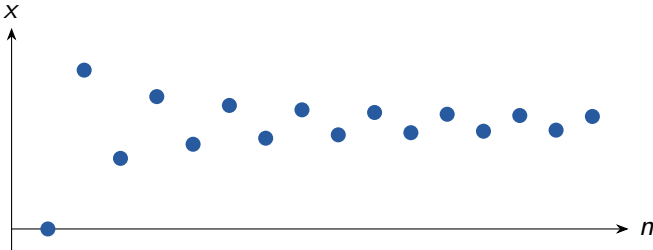
and each x^{n_k} is an element of the original sequence.

Convergent: A sequence (x^n) converges to x if, for every $\varepsilon > 0$, there exists N such that $\rho(x^n, x) < \varepsilon$ for all $n \geq N$.

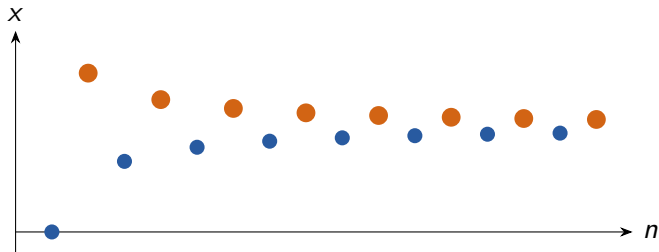
Exercise 1: Sketch of Proof

1. Define a subsequence of any Cauchy that also converges.

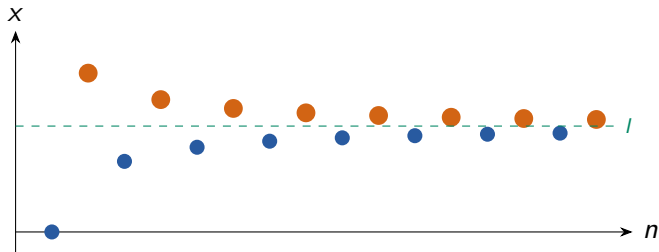
Exercise 1: The even subsequence



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Exercise 2

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Let X be a complete metric space and $Y \subseteq X$ be a subspace of X . Show that if Y is closed in X , then Y is complete.

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$$X \text{ complete, } Y \subseteq X \text{ closed} \implies Y \text{ complete}$$

Exercise 2: Definitions

$$X \text{ complete, } Y \subseteq X \text{ closed} \implies Y \text{ complete}$$

Definitions

Complete: A metric space (X, ρ) is complete if every Cauchy sequence in X converges to a point in X .

Closed: A set $Y \subseteq X$ is closed if every convergent sequence (y^n) in Y converges to a point y in Y .

Exercise 2: Sketch of Proof

1. Take any arbitrary Cauchy x^n in Y .
2. Show that x^n converges.
3. Show that $x^n \rightarrow x \in Y$
4. Conclude that any Cauchy x^n in Y converges in Y .

Exercise 3

Problem

In this exercise, you show that metric spaces $\langle X, \rho \rangle$ which are complete and totally bounded must be sequentially compact. Since X is complete, it suffices to show that every infinite sequence (x_1^n) on X has a subsequence which is Cauchy.

(a) Show that every infinite sequence (x_1^n) in a totally bounded X has an infinite subsequence (x_2^n) which belongs to an open ball with radius $\frac{1}{2}$.

(b) Show that (x_2^n) has an infinite subsequence (x_3^n) which belongs to an open ball of radius $\frac{1}{3}$.

(c) Continuing like this, with open balls of radius $\frac{1}{n}$, define an infinite subsequence $(x_1^1, x_2^2, x_3^3, \dots)$ whose m -th term is the m -th term in (x_m^n) . Show that this sequence is Cauchy.

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In this exercise, you show that metric spaces $\langle X, \rho \rangle$ which are **complete and totally bounded** must be **sequentially compact**.

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Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ Sequentially Compact X

Exercise 3

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Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ **Cauchy subsequence**

Exercise 3: Definitions

Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ Cauchy subsequence

Definitions

Complete: A metric space (X, ρ) is complete if every Cauchy sequence in X converges to a point in X .

Totally bounded: A metric space (X, ρ) is totally bounded if, for every $\varepsilon > 0$, X can be covered by finitely many balls of radius ε .

Sequentially compact: A metric space X is sequentially compact if every sequence in X has a convergent subsequence with limit in X .

Cauchy sequence: A sequence (x^n) is Cauchy if, for every $\varepsilon > 0$, there exists N such that $\rho(x^n, x^m) < \varepsilon$ for all $n, m \geq N$.

Exercise 3: Intuition

Why can we go from

Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ Sequentially Compact X

to

Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ **Cauchy subsequence**?

Exercise 3: Intuition

Why can we go from

Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ Sequentially Compact X

to

Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ **Cauchy subsequence**?

- By **Completeness** we know that Cauchy subsequences converge in X .

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Why can we go from

Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ Sequentially Compact X

to

Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ **Cauchy subsequence**?

- ▶ By **Completeness** we know that Cauchy subsequences converge in X .
- ▶ If each sequence has a Cauchy subsequence which converges, then X is **Sequentially compact**.

Exercise 3: Intuition

Why can we go from

Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ Sequentially Compact X

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Complete and Totally Bounded $\langle X, \rho \rangle \Rightarrow$ **Cauchy subsequence**?

- ▶ By **Completeness** we know that Cauchy subsequences converge in X .
- ▶ If each sequence has a Cauchy subsequence which converges, then X is **Sequentially compact**.

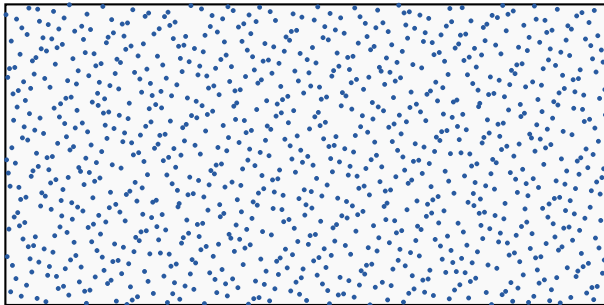
Each sequence has a Cauchy sequence $\xrightarrow{\text{Completeness}}$ Each sequence has a convergent subsequence

Exercise 3: Sketch of Proof

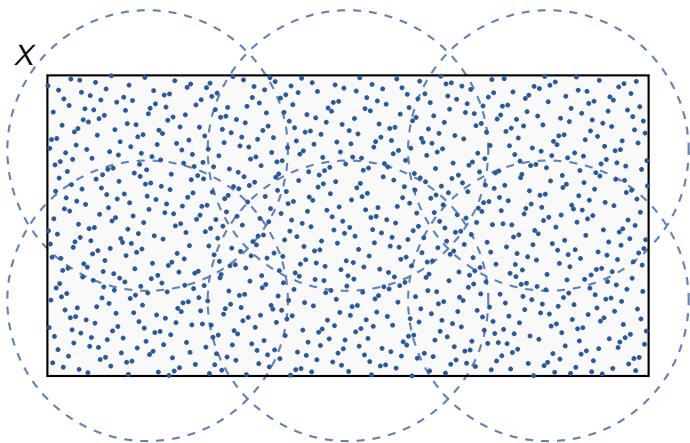
1. Show that every infinite sequence (x_1^n) in a totally bounded X has an infinite subsequence (x_2^n) which belongs to an open ball with radius $\frac{1}{2}$. [▶ Watch on YouTube](#)
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3. Continue like this, with open balls of radius $\frac{1}{n}$.
4. Define an infinite subsequence $(x_1^1, x_2^2, x_3^3, \dots)$ whose m -th term is the m -th term in (x_m^n) .
5. Show that this sequence is Cauchy (the distance between any x_n^n and x_m^m $n, m \geq N$ is at most $2/N$).

Exercise 3: An infinite sequence, finitely many balls

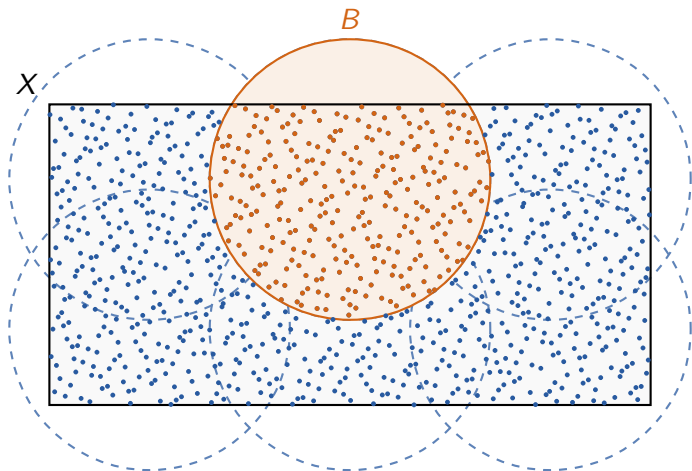
X



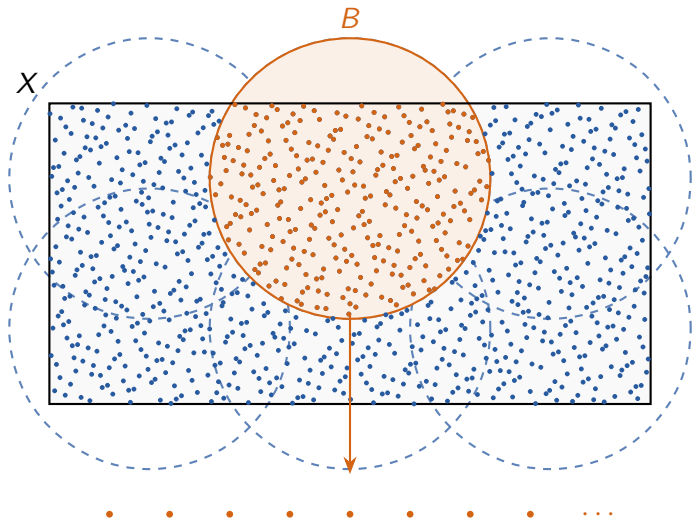
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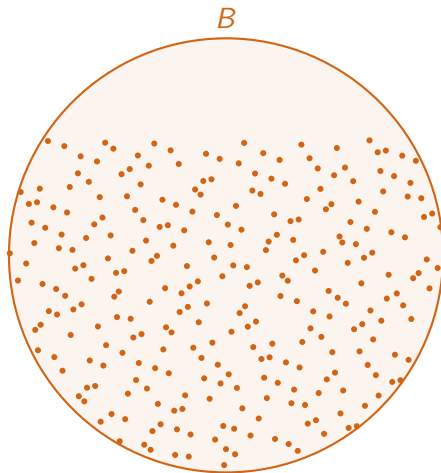
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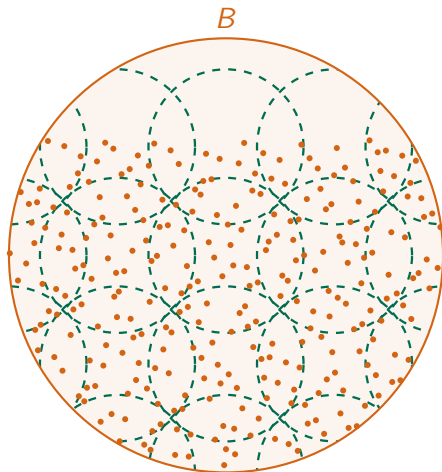
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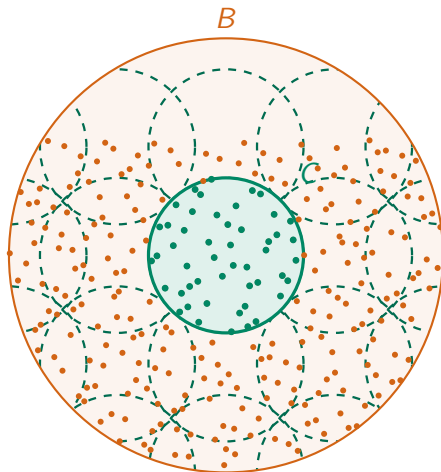
Exercise 3: Zoom into B and repeat



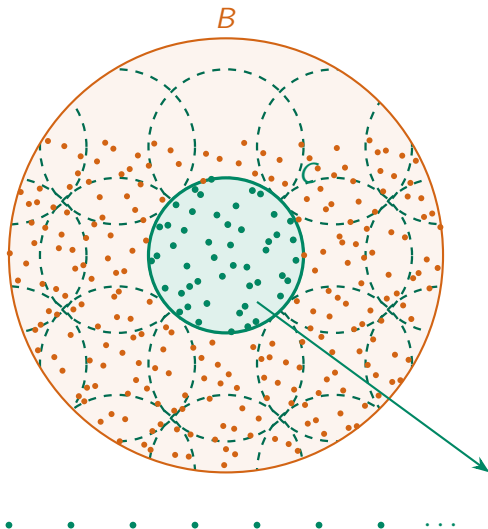
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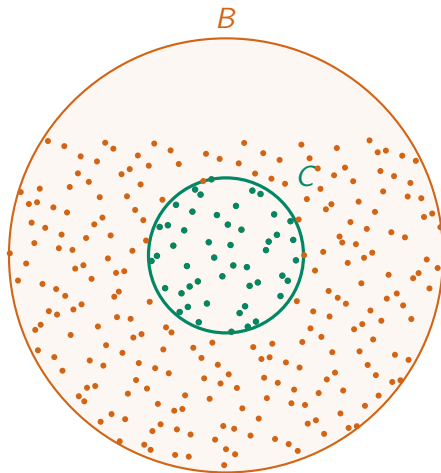
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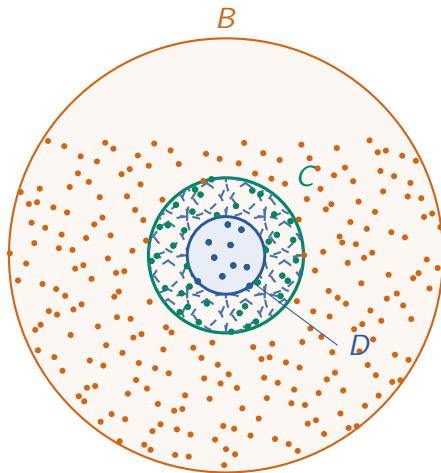
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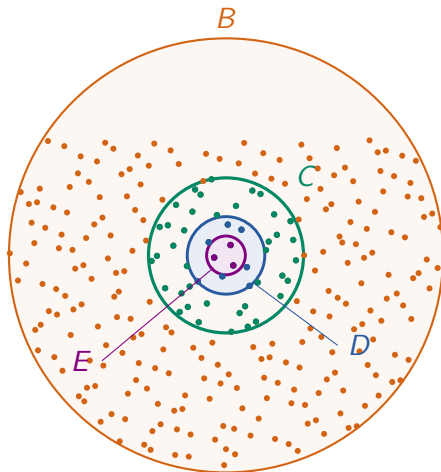
Exercise 3: Repeat inside the selected ball



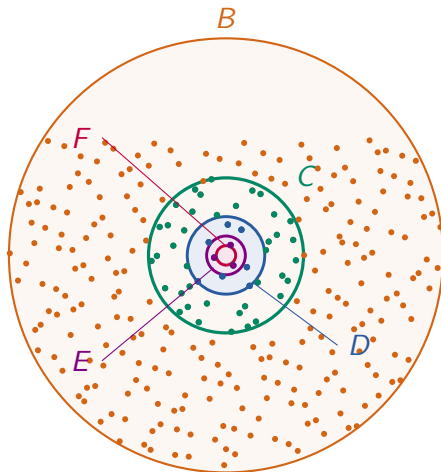
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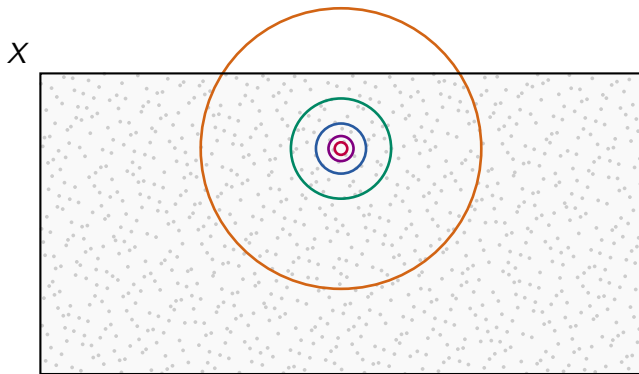
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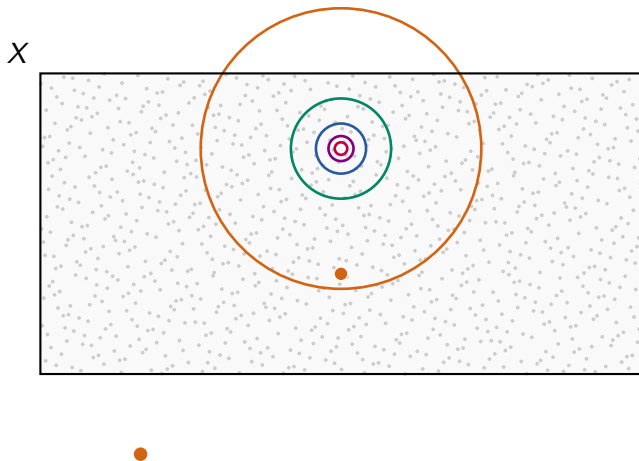
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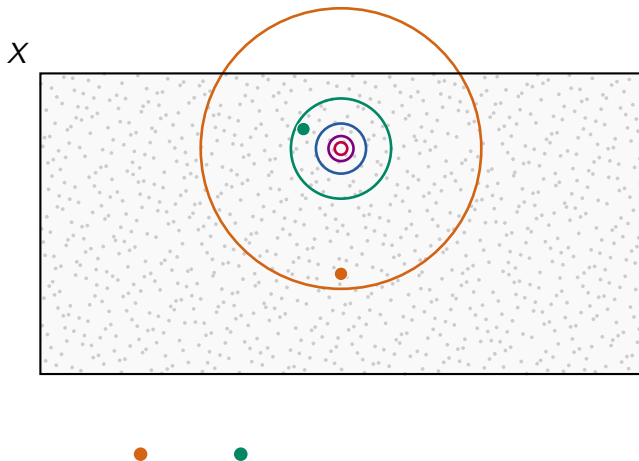
Exercise 3: Back to X , select one term at each stage



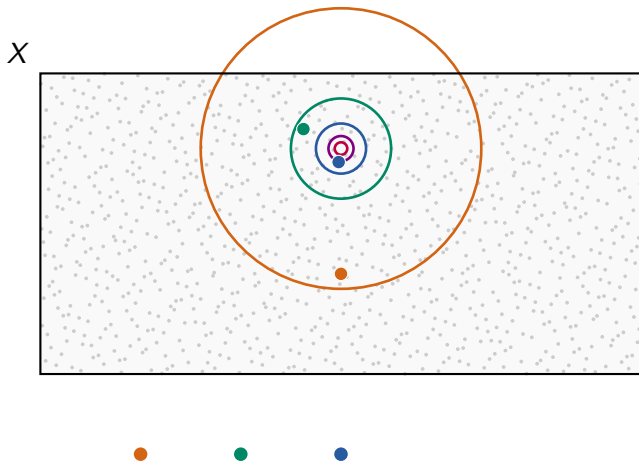
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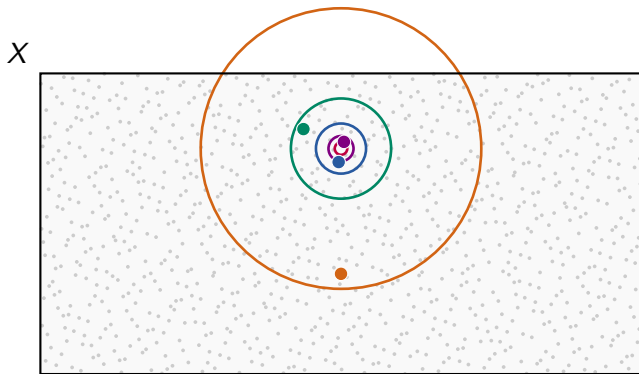
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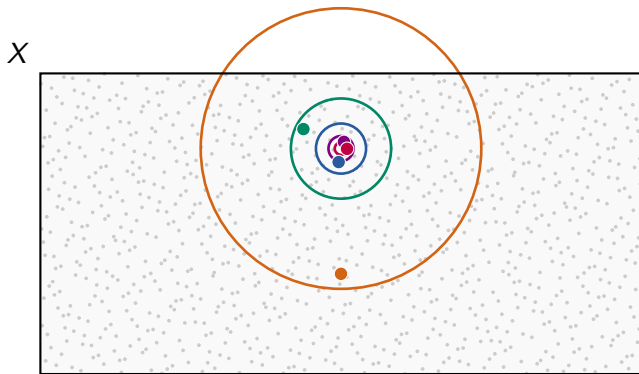
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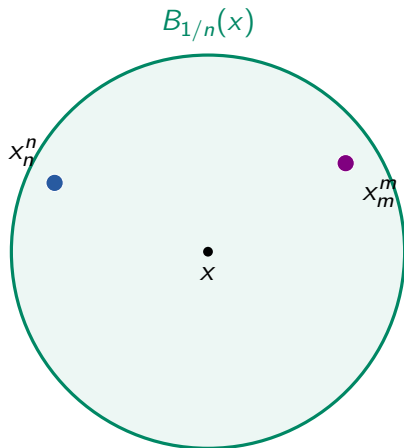
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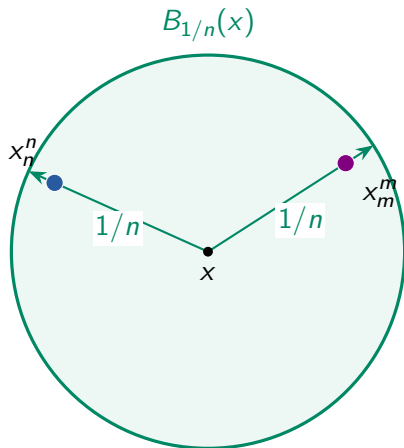
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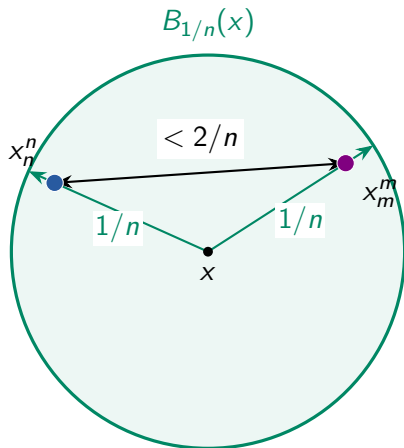
Exercise 3: A small ball controls every later selection



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Exercise 3: How large must N be?

For a target distance $r > 0$, choose an integer N such that

$$\boxed{\frac{2}{N} < r \iff N > \frac{2}{r}}.$$

Target r	Choose N	Check
0.3	7	$2/7 \approx 0.286 < 0.3$

Then for any two diagonal terms with indices $p, q \geq N$,

$$\rho(x_p^p, x_q^q) < \frac{2}{N} < r.$$

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0.1	21	$2/21 \approx 0.0952 < 0.1$
0.01	201	$2/201 \approx 0.00995 < 0.01$

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